

# Digestive integrated pharmacy clinical service

Youna Kang, PharmD; Melanie Cullen, PharmD; Martha Stutsky, PharmD, BCPS; Sera McNutt, PharmD, BCPP; Van Vu, PharmD; Lourdes Cantwell, PharmD, CSP; Kimberly Boothe, PharmD, MHA<sup>1</sup>; Kimhouy Tong, PharmD, BCPS; Vinay Sawant, RPh, MPH, MBA; Marie Renauer, PharmD, MBA, BCACP  
Yale New Haven Health, Department of Pharmacy, New Haven, CT; The Kimber Boothe Group, LLC, Cincinnati, OH<sup>1</sup>

## Background

- Yale New Haven Health is a multi-campus, academic, nonprofit health system that provides high quality care through a variety of ambulatory care service lines and specialty pharmacy services
- Currently, there is limited pharmacist involvement in the management of digestive diseases, including Inflammatory Bowel Disease (IBD) and viral hepatitis
- Digestive Health patients face complex medication regimens requiring close monitoring, counseling, and navigation of prescription procurement<sup>1</sup>
- Medication adherence is imperative given complications, such as disease exacerbations, which can lead to increased healthcare utilization<sup>1</sup>
- A collaborative practice agreement between clinicians and pharmacists in Connecticut legally allows pharmacists to practice collaborative drug therapy management (CDTM) for mutual patients<sup>2</sup>
- Integrated pharmacy services via a CDTM are needed to identify and resolve barriers that lead to increased medication adherence and disease remission

## Objectives

- Create a novel decentralized integrated pharmacy service line for Digestive Health specialty clinics
- Benefits of integrated pharmacy services include the following:
  - Increased patient contact via a referral process for medication management
  - Increased clinical continuity, defined as internal prescription capture
  - Reduced clinician and office administration burden
  - Improved medication access
  - Expanded patient education

## Methods

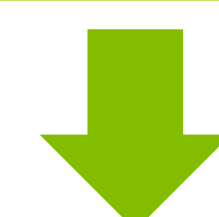
Creation of a comprehensive project plan



Identification of target clinics specializing in Digestive Health



Development and introduction of the service model to clinic providers



Creation and approval of a collaborative practice agreement



Development of a standard operating procedure for workflow outline and pharmacist responsibilities



Education, training, and credentialing of pharmacists

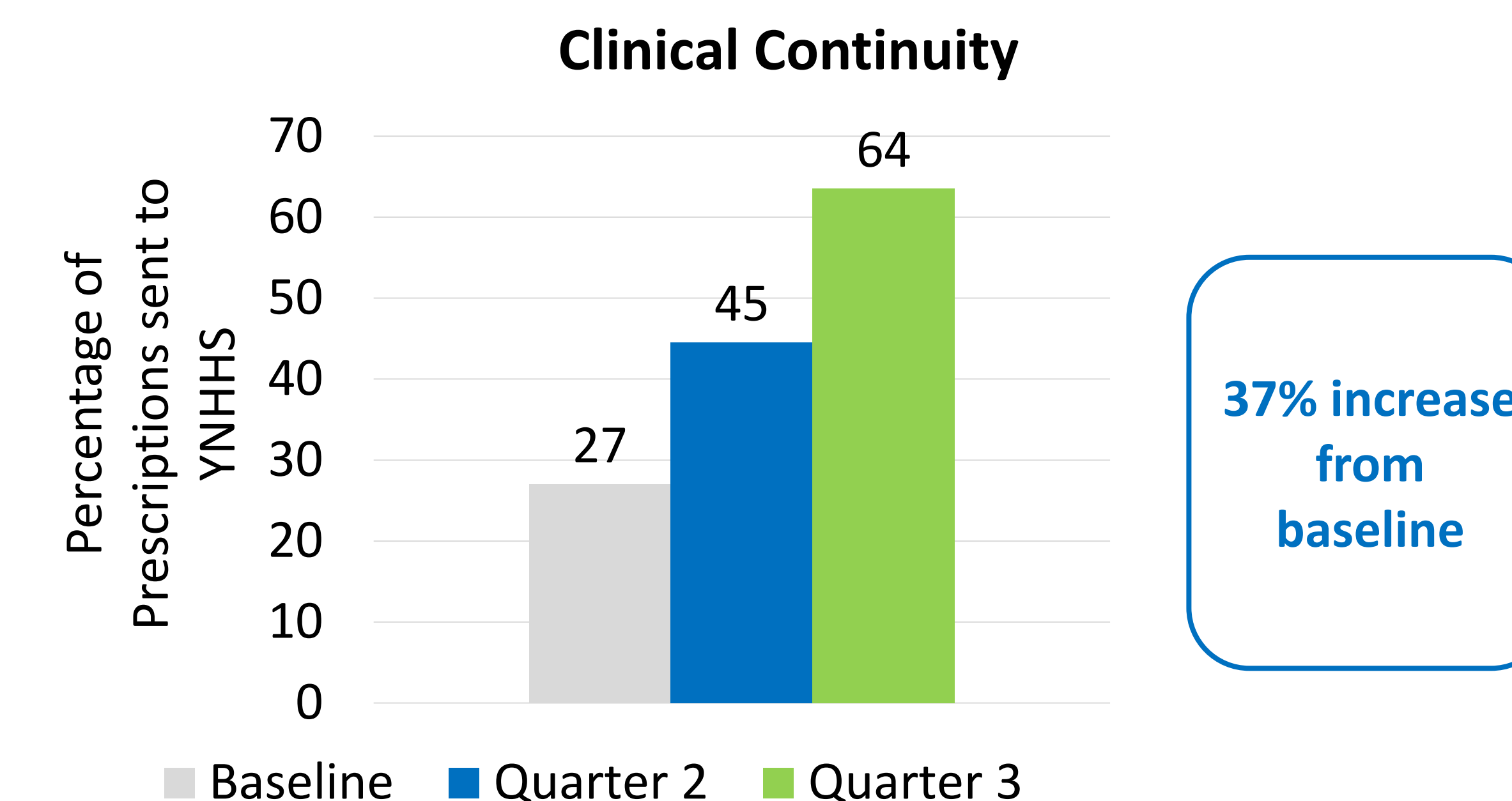
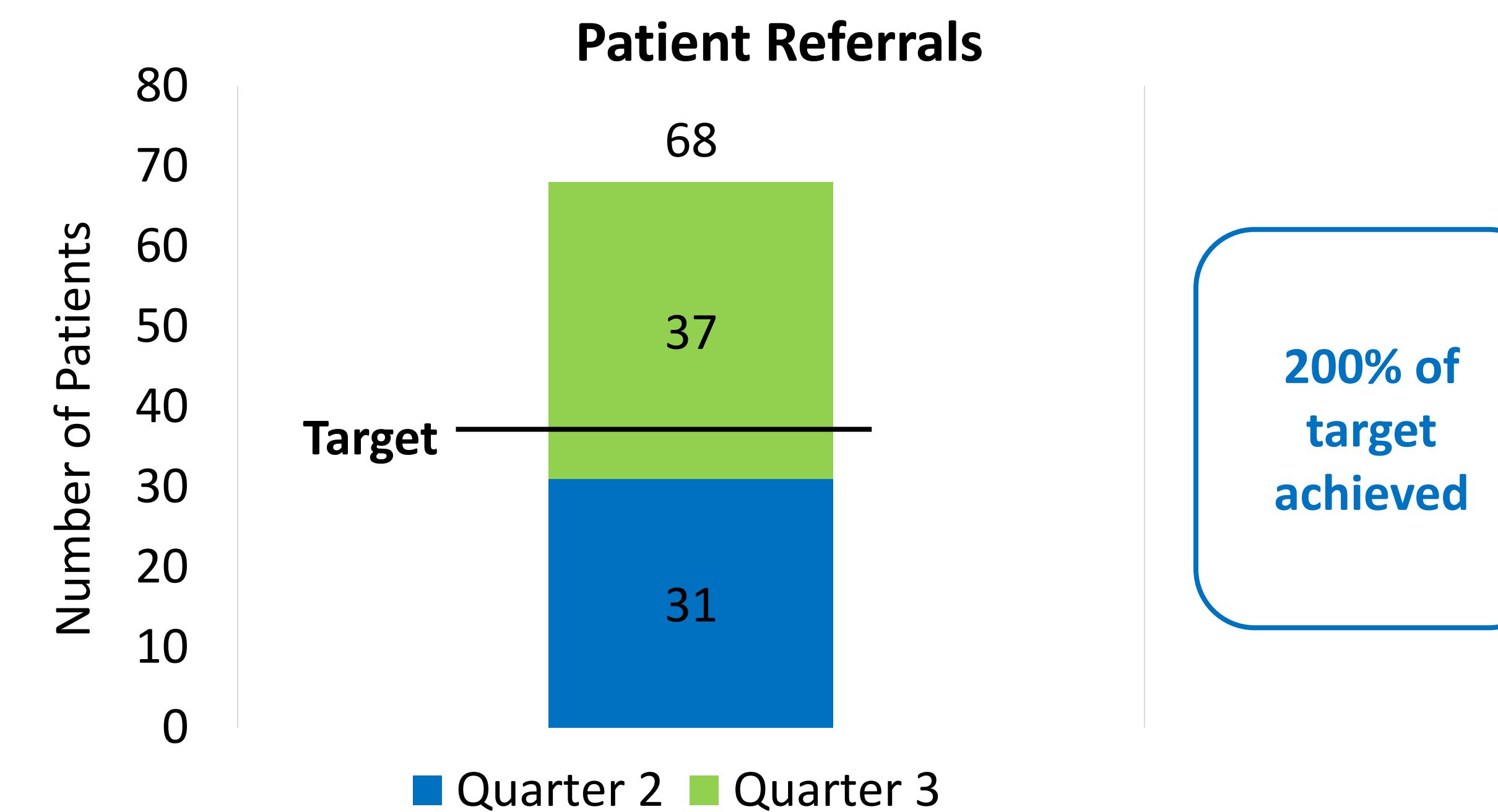


Optimization of the electronic medical record



Measurement of primary outcomes  
(number of patient referrals and clinical continuity)

## Results



## Clinician Testimonials

“Pharmacists’ participation in our patient care has brought us to a **new level of care**. Experience navigating the complicated system of site of care and facility administration lends a new and important degree of **comfort that our patients are being well cared for.**”

“The addition of the clinical pharmacist has been **the best patient care improvement** in my 30 years in practice.”

“Relationship has **eased the burden** on our office staff, enhanced the difficult ordering process on Epic, and offered **tremendous clinical benefit** to our patients.”

## Discussion

- Integration of clinical pharmacy services in identified Digestive Health clinics resulted in patient referrals far exceeding the target as well as clinical continuity increases
- Positive clinician feedback emphasize the impact and value of integrated pharmacy services
- Initial barriers to the implementation of integrated clinical pharmacy services include the ongoing COVID-19 pandemic, stakeholder familiarity with pharmacy services, and the complicated regulatory steps to approve a new collaborative practice agreement
- Evaluation of secondary outcomes including the proportion of days covered and disease-specific clinical outcomes for IBD and viral hepatitis must be completed to provide further evidence of the clinical impact of the integrated pharmacist

## Conclusions

- A novel integrated service model was successfully implemented in three clinic sites for Digestive Health
- The role of the integrated pharmacist has led to increased clinical continuity, patient referrals, and Digestive Health clinician satisfaction

## Future Directions

- Continued service optimization based on pharmacist, clinician, and patient feedback
- Evaluation of disease-specific clinical outcomes and laboratory monitoring adherence
- CDTM updates to include supportive care such as vitamin supplementation and drug level monitoring

## References

- Tiao DK, Chan W, Jeganathan J, Chan JT, Perry J, Selinger CP, et al. Inflammatory bowel disease pharmacist adherence counseling improves medication adherence in Crohn’s disease and ulcerative colitis. *Inflamm Bowel Dis*. 2017;23:1257-1261.
- Collaborative Drug Therapy Management, R.C.S.A. §§ 20-631-1— 20-631-3 (2015).